

Module A: Pseudoinverse for “solve $Ax = b$ in general” (Ch. 12.5) — SVD view

Motivation. When $A \in \mathbb{R}^{m \times n}$ is not invertible (rectangular or rank-deficient), we still want a *canonical* solution. A natural choice is:

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2 \quad \text{and among all minimizers pick the one with minimum } \|x\|.$$

This yields a unique answer $x^* = A^\dagger b$ (Moore–Penrose pseudoinverse).

SVD cheat sheet. Let $\text{rank}(A) = r$. Then there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and

$$A = U\Sigma V^\top, \quad \Sigma = \begin{bmatrix} \text{diag}(\sigma_1, \dots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix},$$

$$\Sigma^\dagger = \begin{bmatrix} \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}) & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{n \times m},$$

where $\sigma_1 \geq \dots \geq \sigma_r > 0$.

and

$$A^\dagger := V\Sigma^\dagger U^\top.$$

Geometric picture:

$x \xrightarrow{V^\top} \text{rotate} \xrightarrow{\Sigma} \begin{matrix} \text{stretch on } r \\ \text{singular dirs,} \\ \text{kill nullspace} \end{matrix} \xrightarrow{U} \text{rotate}$

Then $x^* := A^\dagger b$ is the unique vector that:

- solves $\min_x \|Ax - b\|$;
- among all least-squares minimizers, has minimum $\|x\|$.

Sanity check (special cases).

- Full column rank ($m \geq n$, $\text{rank}(A) = n$): $A^\dagger = (A^\top A)^{-1} A^\top$.
- Full row rank ($m \leq n$, $\text{rank}(A) = m$): $A^\dagger = A^\top (AA^\top)^{-1}$.

Remarks

- $Ax = b$ may be inconsistent. Then $A^\dagger b$ is **not** an exact solution; it is the **minimum-norm least-squares** solution.
- SVD view: components of b in $\mathcal{N}(A^\top)$ *cannot be fitted*; the best you can do is match the projection onto $\mathcal{R}(A)$.

Practice: Textbook 12.29 (full-rank factorization reduction)

Consider $\min_{x \in \mathbb{R}^n} \|Ax - b\|^2$ where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. Let x^* be a solution. Suppose $A = BC$ is a full-rank factorization: $\text{rank } A = \text{rank } B = \text{rank } C = r$, with $B \in \mathbb{R}^{m \times r}$, $C \in \mathbb{R}^{r \times n}$. Show that the minimizer of $\|By - b\|$ over $\mathbb{R}^r$ is Cx^* .

Answer: Let $f(x) = \|BCx - b\|^2$ and $g(y) = \|By - b\|^2$. Note $f(x) = g(Cx)$. Since x^* minimizes f , the normal equation gives

$$\nabla f(x^*) = 2C^\top B^\top (BCx^* - b) = 0 \quad \Rightarrow \quad C^\top w = 0, \quad \text{where } w := B^\top (BCx^* - b) \in \mathbb{R}^r.$$

Because $\text{rank}(C) = r$ (full row rank), the map $C^\top : \mathbb{R}^r \rightarrow \mathbb{R}^n$ has full column rank and is injective, so $C^\top w = 0 \Rightarrow w = 0$. Hence

$$B^\top (BCx^* - b) = 0.$$

Let $y^* := Cx^*$. Then $B^\top (By^* - b) = 0$, which is exactly the normal equation for minimizing $g(y)$. Also $\text{rank}(B) = r$ (full column rank) implies $B^\top B \succ 0$, so g has a *unique* minimizer. Therefore $y^* = Cx^*$ is the minimizer of $\|By - b\|$ over $\mathbb{R}^r$.



What is an LP? A *linear program* optimizes a linear objective over linear constraints, i.e. over a polyhedron. Typical forms:

$$\min c^\top x \quad \text{s.t. } Ax \leq b, \quad x \geq 0$$

$$\max c^\top x \quad \text{s.t. } Ax = b, \quad x \text{ free.}$$

Standard form (the one simplex likes).

$$\min c^\top x \quad \text{s.t. } Ax = b, \quad x \geq 0,$$

with $A \in \mathbb{R}^{m \times n}$, $m < n$, $\text{rank}(A) = m$, and w.l.o.g. $b \geq 0$.

Basic solution / BFS . Pick m linearly independent columns of A to form a basis matrix $B \in \mathbb{R}^{m \times m}$, reorder columns so $A = [B, D]$. Set *nonbasic* variables to 0, solve

$$Bx_B = b, \quad x = \begin{bmatrix} x_B \\ 0 \end{bmatrix}.$$

If additionally $x \geq 0$, it is a **basic feasible solution (BFS)**. If some basic components are 0, the BFS is **degenerate**.

Conversion rules (toolkit).

- **Max $\rightarrow$ Min:** $\max c^\top x \iff \min(-c)^\top x$.
- **$\leq$ constraint:** $a^\top x \leq \beta \iff a^\top x + s = \beta, \quad s \geq 0$ (slack).
- **$\geq$ constraint:** $a^\top x \geq \beta \iff a^\top x - s = \beta, \quad s \geq 0$ (surplus).
- **Free variable:** $x_i \in \mathbb{R} \iff x_i = x_i^+ - x_i^-, \quad x_i^\pm \geq 0$.
- **Absolute value:** $|x_i| \leq t \iff -t \leq x_i \leq t$ (then use slack), or split $x_i = x_i^+ - x_i^-$.

Practice: Textbook 15.3 (convert ℓ_1 objective to standard form LP)

Consider

$$\text{minimize } c_1|x_1| + c_2|x_2| + \dots + c_n|x_n| \quad \text{subject to } Ax = b,$$

where $c_i \neq 0$ for $i = 1, \dots, n$. Convert this into an equivalent *standard form* LP.

Answer: Introduce for each i the split variables $x_i^+, x_i^- \geq 0$ such that

$$x_i = x_i^+ - x_i^-, \quad |x_i| = x_i^+ + x_i^-.$$

Let $x^+ = (x_1^+, \dots, x_n^+)^\top$, $x^- = (x_1^-, \dots, x_n^-)^\top$, and define the stacked variable $z = \begin{bmatrix} x^+ \\ x^- \end{bmatrix} \in \mathbb{R}^{2n}$ with $z \geq 0$.

Then the constraint becomes

$$A(x^+ - x^-) = b \iff \underbrace{[A, -A]}_{\tilde{A}} z = b.$$

The objective becomes

$$\sum_{i=1}^n c_i(x_i^+ + x_i^-) = \underbrace{[c^\top, c^\top]}_{\tilde{c}^\top} z, \quad \text{where } c = (c_1, \dots, c_n)^\top.$$

Hence an equivalent standard form LP is:

$$\min \tilde{c}^\top z \quad \text{s.t. } \tilde{A}z = b, \quad z \geq 0,$$

with $\tilde{A} = [A, -A]$ and $\tilde{c} = [c; c]$.